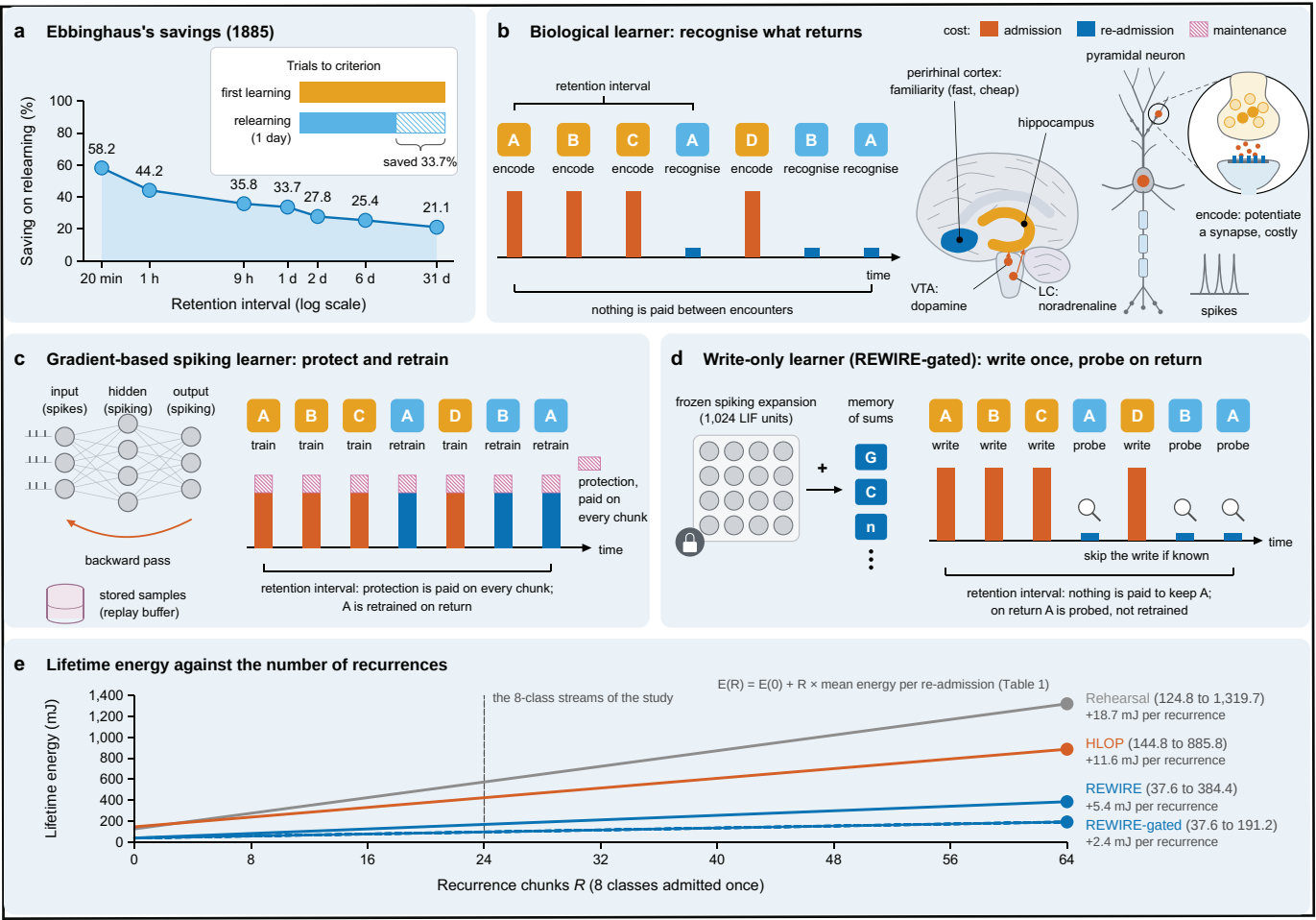

a
Ebbinghaus's savings (1885)
Trials to criterion
first learning
relearning (1 day)
saved 33.7%
Saving on relearning (%)
100
80
60
40
20
0
58.2
44.2
35.8
33.7
27.8
25.4
21.1
20 min
1 h
9 h
1 d
2 d
6 d
31 d
Retention interval (log scale)
b
Biological learner: recognise what returns
cost: admission re-admission maintenance
retention interval
A B C A D B A
encode encode encode recognise encode recognise recognise
time
nothing is paid between encounters
perirhinal cortex: familiarity (fast, cheap)
hippocampus
VTA: dopamine
LC: noradrenaline
pyramidal neuron
encode: potentiate a synapse, costly
spikes
c
Gradient-based spiking learner: protect and retrain
input (spikes)
hidden (spiking)
output (spiking)
backward pass
stored samples (replay buffer)
A B C A D B A
train train train retrain train retrain retrain
protection, paid on every chunk
time
retention interval: protection is paid on every chunk; A is retrained on return
d
Write-only learner (REWIRE-gated): write once, probe on return
frozen spiking expansion (1,024 LIF units)
memory of sums
G
C
n
A B C A D B A
write write write probe write probe probe
time
skip the write if known
retention interval: nothing is paid to keep A; on return A is probed, not retrained
e
Lifetime energy against the number of recurrences
E(R) = E(0) + R × mean energy per re-admission (Table 1)
the 8-class streams of the study
Lifetime energy (mJ)
1,400
1,200
1,000
800
600
400
200
0
0
8
16
24
32
40
48
56
64
Recurrence chunks R (8 classes admitted once)
Rehearsal (124.8 to 1,319.7)
+18.7 mJ per recurrence
HLOP (144.8 to 885.8)
+11.6 mJ per recurrence
REWIRE (37.6 to 384.4)
+5.4 mJ per recurrence
REWIRE-gated (37.6 to 191.2)
+2.4 mJ per recurrence